

Supplementary Examination Summer 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: CV501 Design of Steel Structure

Max Marks: 60

Date: 07/08/2023

Duration: 03.00 Hrs.

Instructions to the Students:

1. Solve any five questions from Q. no. 1 to Q. no 6 .
1. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.
4. Use of IS:800-1984 and 2007, Steel table and IS:875-1987 is allowed

(Level/CO) Marks

Q. 1 Solve Any Two of the following

- A) Differentiate between the welded connection and riveted connection (CO1) 6
- B) A 6 mm thick angle section is jointed to a 10 mm thick gusset plate. The angle is supporting a load of 60kN. Find out the number of 20 mm diameter power driven shop rivets (CO2) 6
- C) A single riveted double cover butt joint is used to connect two plates of 10 mm thick with chain riveting. The rivets used are power driven 16 mm diameter at a pitch of 60 mm. find out safe load per pitch length and efficiency of the joint. ($f_y=250\text{Mpa}$) 6

Q. 2 Solve the following Questions

- A) Give the advantages of steel as a structural material (CO1) 6
- B) Determine the strength of a 20mm dia. ordinary turned bolt if it is used to connect (CO3) 6
- a. Two plates each of 10mm thick
- b. Three plates, the outer two plates 8mm thick each and the intermediate plate 14mm thick

If shear strength of rivet is 100 Mpa and bearing strength is 300 Mpa

Q. 3 Solve Any Two of the following.

- A) Explain in detail- Eccentric connection and its types (CO3) 6
- B) Determine the tensile strength of a roof truss diagonal member (CO1) 6
- 100mm*75mm*10mm ($F_y=260\text{Mpa}$) connected to the gusset plate by 20mm diameter power driven rivets in one row along the length of the member. The short leg of the angle is kept outstanding.
- C) Explain in detail- Lug angle (CO1) 6
- Tension Splices

- Q.4** Design a beam of 4m effective span carrying a uniformly distributed load of 15 kN/m, if the compression flange is laterally unsupported. ($f_y=250\text{Mpa}$) assume dead load = 500 N/m. **(CO3)** **12**
- Q. 5** For a roof truss system of an industrial building following data is given, **(CO3)** **12**
- a) Span of truss= 6m,
 - b) Spacing of truss= 3m centre to centre.
 - c) Inclination of roof= 30°
 - d) Spacing of purlin= 2m centre to centre.
 - e) Wind pressure= 1 kN/m
 - f) Roof coverage= A. C. sheets weighing 150 N/m.
- Design a I-section purlin.
- Q. 6 Solve Any Two of the following.**
- A)** Differentiate between elastic design and plastic design. **(CO4)** **6**
 - B)** Explain **(CO4)** **6**
 - Slab base
 - Gusseted base
 - Grillage foundation
 - C)** Explain in detail: limit state of strength and limit state of serviceability **(CO4)** **6**